

Introduction to Machine Learning | Numerical Practice Set

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Instructions: Show all calculation steps. Use suitable formulas wherever required. Round final values to two decimal places unless an exact value is requested.

Section A : Data Preparation and Feature Processing

1. Train test split: A dataset has 2,500 records. Create an 80:20 train test split. How many records will be in the training and testing sets?
2. Stratified split: A dataset has 1,200 students: 900 Pass and 300 Fail. Using a stratified 80:20 split, estimate the number of Pass and Fail records in both training and testing sets.
3. Feature count: A dataset has 50 columns. After removing `student_id`, `name`, `teacher_id`, `final_grade_letter`, and the target column, how many input features remain?
4. Descriptive statistics: The `attendance_percentage` values are 65, 70, 75, 80, and 85. Calculate the mean and median.
5. Mean imputation: The marks of five students are 40, 50, 60, 70, and 80. One value is missing. Calculate the mean-imputed value.
6. Mean versus median: The weekly study hours are 2, 3, 4, 5, and 50. Calculate the mean and median. Which imputation value would be more appropriate for a missing observation, and why?
7. IQR calculation: For a feature, $Q1 = 20$ and $Q3 = 40$. Calculate the IQR, lower bound, and upper bound using the IQR method.
8. Outlier identification: Using the bounds calculated in Question 7, identify possible outliers from these values: 5, 18, 22, 35, 39, 45, and 72.
9. Feature engineering: A student has Test 1 = 70, Test 2 = 80, and Final Exam = 90. Calculate the engineered feature `average_score`.
10. Standardization: A numerical feature has training mean $\mu = 50$ and standard deviation $\sigma = 10$. Standardize the value $x = 70$.
11. Min max normalization: A feature has minimum value 20 and maximum value 120. Calculate the Min Max normalized value of $x = 45$.
12. Normalization values: A feature has values 10, 20, 30, 40, and 50. After Min Max normalization, what are the transformed values for 10, 30, and 50?
13. One hot encoding: A training dataset has categorical feature `subject` with four categories: Mathematics, Physics, Chemistry, and English. How many new columns will one hot encoding create when `drop_first = True`?

14. Duplicate records: A dataset contains 600 records. It has 24 duplicate rows. What will be the dataset size after removing exact duplicates?
15. Missing percentage: A dataset has 1,000 records, and 80 contain missing values in `study_hours_per_week`. Calculate the missing-value percentage.

Section B : Model Evaluation and Validation

16. Classification metrics: A binary classifier gives $TP = 72$, $TN = 108$, $FP = 12$, and $FN = 8$. Calculate accuracy, precision, recall, and F1-score.
17. Classification metrics and decision: A disease-prediction model has $TP = 45$, $FP = 5$, $FN = 15$, and $TN = 135$. Calculate precision and recall. Which metric should be prioritised if missing a disease case is dangerous?
18. Regression metrics: Actual marks are [60, 70, 80, 90], while predicted marks are [65, 68, 82, 85]. Calculate MAE, MSE, and RMSE.
19. Residuals: Actual target values are [50, 60, 70], and model predictions are [55, 58, 72]. Calculate the residual for each observation using $\text{residual} = \text{actual} - \text{predicted}$.
20. Overfitting: A model achieves training accuracy of 98 percent and test accuracy of 72 percent. What issue does this suggest? State one method to reduce it.
21. Cross validation: In a 5-fold cross-validation experiment on 500 records, how many records are used for validation in each fold, and how many are used for training in each fold?
22. Correlation: A correlation coefficient between attendance and final exam score is $r = 0.82$. Interpret its direction and strength. Does this prove that attendance causes high marks?
23. Pass rates: The pass rates for three subjects are Mathematics = 80 percent, Physics = 65 percent, and Chemistry = 75 percent. If each subject has 200 students, calculate the number of students who passed each subject.
24. Final feature count: A feature matrix initially has 8 numerical features and 3 categorical features. The categorical variables have 2, 4, and 5 categories respectively. After one hot encoding with `drop_first = True`, how many total features will the model receive?
25. Data leakage: A model uses the whole dataset to calculate mean attendance before splitting into train and test sets. Name this error and explain why it can make the final score unreliable.